

Lab 1: Preflight & Environment Readiness

Course: IT4065C – Data Technologies Administration

Repository: <https://github.com/ntious/IT4065C-Labs>

Time Required: ≤ 1 hour

Environment: Individual Ubuntu VM (VMware Cloud Services)

Goal of Lab 1

By the end of this lab, you will be able to:

1. Log into your Ubuntu virtual machine
2. Clone the course GitHub repository
3. Install PostgreSQL locally on your VM
4. Install dbt for PostgreSQL
5. Configure dbt to connect to a local database
6. Run dbt debug successfully
7. Run basic SQL queries against a local database

Note:

No transformations, no modeling, and no dbt run are required in this lab.

Step 1: Choose Your Ubuntu Environment (Required)

In **Step 1**, you must choose **one** of the following options:

- **Option A:** Use the UC-provided Ubuntu Virtual Machine (Sandbox)
- **Option B:** Install Ubuntu on your own Windows machine using the Microsoft Store

👉 You only need to complete **ONE** option. Do not do both.

Step 1.1 (OptionA): Log into Your Ubuntu VM (Using the UC Sandbox)

1. Log into your UC Sandbox Ubuntu VM at:
<https://range.ocri.io>
2. Use the following credentials (UC Sandbox only):
 - **Ubuntu VM Password:** xxxxxx
3. Once logged in, open a Terminal window.
4. Confirm the operating system by running:
lsb_release -a
5. You should see Ubuntu 22.04 LTS (or higher)



```
student@gozer-ubuntu:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.4 LTS
Release:       22.04
Codename:      focal
student@gozer-ubuntu:~$
```

⚠ If the VM does not load correctly or you cannot log in, stop and contact your instructor or IT support before proceeding.

Step 1.2 (Option B): Creating Your Own Ubuntu Environment on Windows (Microsoft Store)

In this step, you will install Ubuntu Linux on Windows using the Microsoft Store. This creates a Linux environment using Windows Subsystem for Linux (WSL), no full virtual machine is required.

⚠ This option is required only if you are not using the UC Sandbox VM.

Requirements

Before you begin, make sure you have:

- Windows 10 (version 21H2 or later) or Windows 11
- Administrator access on your computer
- Internet connection

Step 1.2.1: Open the Microsoft Store

1. Click the Start menu
2. Type Microsoft Store
3. Open the Microsoft Store app

Step 1.2.2: Download Ubuntu

1. In the Microsoft Store search bar, type Ubuntu
2. Select Ubuntu 22.04 LTS
 - ⚠ Do not select Ubuntu Server or preview versions
3. Click Get or Install
4. Wait for the download to complete

Step 1.2.3: Launch Ubuntu

1. Once installation finishes, click Open
(or open Ubuntu from the Start menu)

Ubuntu will initialize for the first time. This may take a few minutes.

Step 1.2.4: Create Your Linux User Account

When prompted in the Ubuntu window:

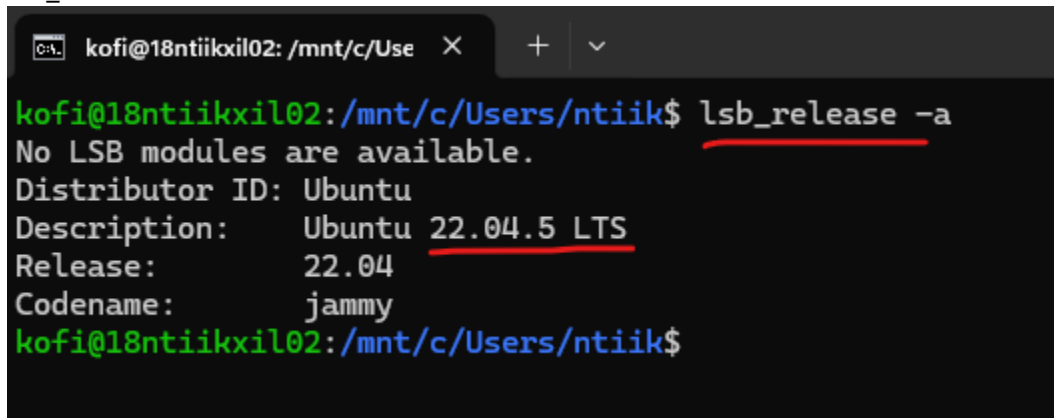
1. Enter a Linux username
 - Lowercase letters only (e.g., student, ubuntu)
2. Enter a password
 - You will not see characters as you type; this is normal
3. Confirm the password

⚠ Remember this password. You will need it throughout the course.

Step 1.2.5: Verify Ubuntu Installation

Once you see the Linux prompt, run:

`lsb_release -a`



```
kofi@18ntiikxil02: /mnt/c/Use X + v
kofi@18ntiikxil02:/mnt/c/Users/ntiik$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.5 LTS
Release:        22.04
Codename:       jammy
kofi@18ntiikxil02:/mnt/c/Users/ntiik$
```

Step 2: Update System Packages (Required for All Students)

Regardless of which option you chose, run the following commands **once**:

1. Run the following commands once:

`sudo apt update`

`sudo apt upgrade -y`

This helps prevent package conflicts later.

Step 3: Install Required System Tools

1. Install Git, Python, and PostgreSQL:

`sudo apt install -y git python3 python3-pip python3-venv postgresql postgresql-contrib unzip nano wget`

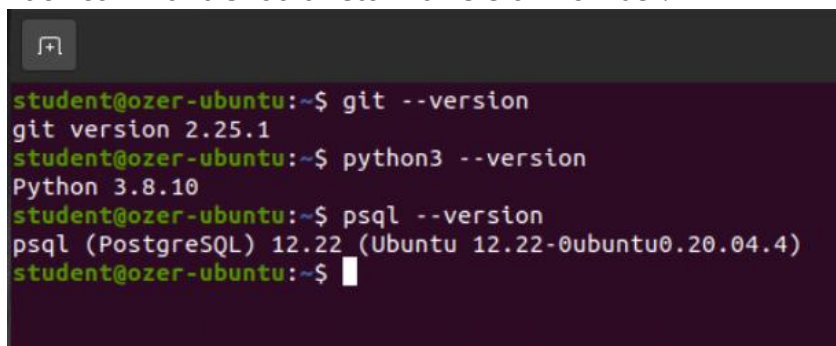
2. Verify each installation:

`git --version`

`python3 --version`

`psql --version`

Each command should return a version number.

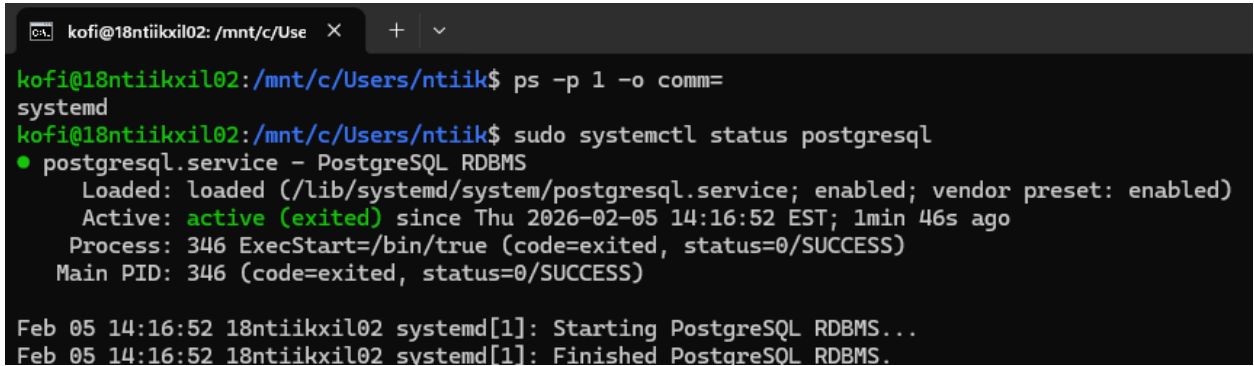


```
student@ozer-ubuntu:~$ git --version
git version 2.25.1
student@ozer-ubuntu:~$ python3 --version
Python 3.8.10
student@ozer-ubuntu:~$ psql --version
psql (PostgreSQL) 12.22 (Ubuntu 12.22-0ubuntu0.20.04.4)
student@ozer-ubuntu:~$
```

Step 4: Create Database and Schemas

4.1 Check PostgreSQL service status

1. Run this:
`ps -p 1 -o comm=`
2. If the output is **systemd**, run:
`sudo systemctl status postgresql`



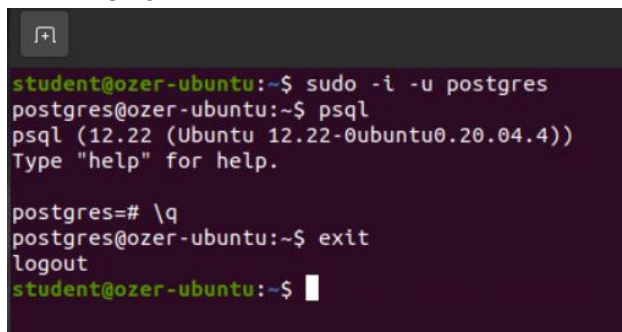
```
kofi@18ntiikxil02: /mnt/c/Use X + v
kofi@18ntiikxil02:/mnt/c/Users/ntiik$ ps -p 1 -o comm=
systemd
kofi@18ntiikxil02:/mnt/c/Users/ntiik$ sudo systemctl status postgresql
● postgresql.service - PostgreSQL RDBMS
   Loaded: loaded (/lib/systemd/system/postgresql.service; enabled; vendor preset: enabled)
   Active: active (exited) since Thu 2026-02-05 14:16:52 EST; 1min 46s ago
     Process: 346 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
    Main PID: 346 (code=exited, status=0/SUCCESS)

Feb 05 14:16:52 18ntiikxil02 systemd[1]: Starting PostgreSQL RDBMS...
Feb 05 14:16:52 18ntiikxil02 systemd[1]: Finished PostgreSQL RDBMS.
```

3. If the output is **not systemd** (common in WSL), run:
`sudo service postgresql status`
4. If PostgreSQL is NOT running, start it:
 - If systemd is available:
`sudo systemctl start postgresql`
 - If systemd is NOT available:
`sudo service postgresql start`

4.2 Switch to the postgres user & open the PostgreSQL shell

1. Switch to the postgres user:
`sudo -i -u postgres`
2. Open the PostgreSQL shell:
`Psq`
3. Exit PostgreSQL:
`\q`
4. Exit the postgres user:
`exit`



```
student@ozer-ubuntu:~$ sudo -i -u postgres
postgres@ozer-ubuntu:~$ psql
psql (12.22 (Ubuntu 12.22-0ubuntu0.20.04.4))
Type "help" for help.

postgres=# \q
postgres@ozer-ubuntu:~$ exit
logout
student@ozer-ubuntu:~$
```

Step 5: Create Your Local Course Database and Schema

1. From your normal Ubuntu user account, run:

```
sudo -u postgres createdb it4065c || echo "Database already exists, continuing..."
```

If you see "database already exists," this is expected. Continue to the next step.

2. Set postgres password so dbt can authenticate over localhost TCP

```
sudo -u postgres psql -d postgres -c "ALTER USER postgres WITH PASSWORD  
'Pa$$w0rd123!';"
```

3. Open the course database in psql (local socket)

```
sudo -u postgres psql it4065c
```

4. Inside PostgreSQL, run:

```
CREATE SCHEMA raw;
```

```
CREATE SCHEMA student_kofi;
```

5. \q

This simulates schema-level governance isolation used later in the course.

```
student@ozer-ubuntu:~$ sudo -u postgres createdb it4065c || echo "Database already exists, continuing..."
student@ozer-ubuntu:~$ sudo -u postgres psql it4065c
psql (12.22 (Ubuntu 12.22-0ubuntu0.20.04.4))
Type "help" for help.

it4065c=# CREATE SCHEMA raw;
CREATE SCHEMA
it4065c=# CREATE SCHEMA student_kofi;
CREATE SCHEMA
it4065c=# \q
student@ozer-ubuntu:~$
```

4. Verify Schema Creation (Optional but Recommended)

```
sudo -u postgres psql it4065c -c "\dn+"
```

```
student@ozer-ubuntu:~$ sudo -u postgres psql it4065c -c "\dn+"
List of schemas
-----+-----+-----+-----+
Name      | Owner  | Access privileges | Description
-----+-----+-----+-----+
public    | postgres | postgres=UC/postgres+ | standard public schema
raw       | postgres |                     |
student_kofi | postgres |                     |
(3 rows)

student@ozer-ubuntu:~$
```

Step 6: Clone the Course GitHub Repository

Step 6A: Preferred (Git clone)

1. From your home directory, run:

```
git clone https://github.com/ntious/IT4065C-Labs.git  
cd IT4065C-Labs
```
2. Confirm the repository structure:

```
ls
```

If git clone fails (timeout, network error, TLS/cert error), **use Step 6B (ZIP method)** below and continue.

Step 6B: If cloning fails (Download ZIP and unzip)

Add a clean, copy-paste fallback that:

1. Downloads the ZIP
2. Unzips it
3. Renames it to IT4065C-Labs (so your later steps remain correct)
4. Verifies folder structure

Run all the following commands:

```
cd ~  
sudo apt install -y unzip  
wget -O IT4065C-Labs.zip https://github.com/ntious/IT4065C-Labs/archive/refs/heads/main.zip  
unzip -q IT4065C-Labs.zip  
rm -f IT4065C-Labs.zip  
rm -rf IT4065C-Labs  
mv IT4065C-Labs-main IT4065C-Labs  
cd IT4065C-Labs  
ls
```

You should see the following folders (either step 6A or 6B):

- docs
- labs
- dbt
- scripts
- submissions

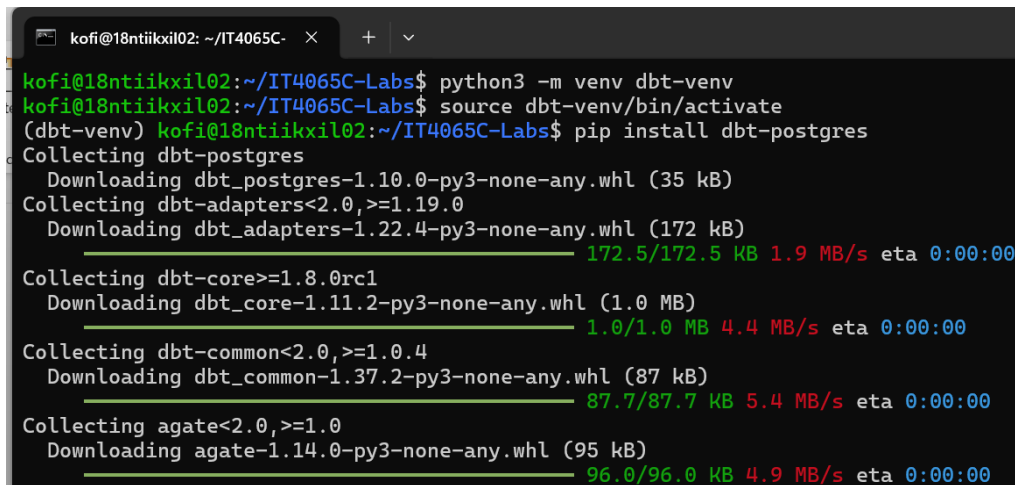
 **Screenshot 1:** Take a screenshot showing the repository structure (docs, labs, dbt).

```
kofi@18ntiikxil02:~$ git clone https://github.com/ntious/IT4065C-Labs.git  
Cloning into 'IT4065C-Labs'...  
remote: Enumerating objects: 77, done.  
remote: Counting objects: 100% (77/77), done.  
remote: Compressing objects: 100% (67/67), done.  
remote: Total 77 (delta 23), reused 9 (delta 2), pack-reused 0 (from 0)  
Receiving objects: 100% (77/77), 23.24 KiB | 1.11 MiB/s, done.  
Resolving deltas: 100% (23/23), done.  
kofi@18ntiikxil02:~$ cd IT4065C-Labs  
kofi@18ntiikxil02:~/IT4065C-Labs$ ls  
LICENSE README.md dbt docs labs scripts submissions  
kofi@18ntiikxil02:~/IT4065C-Labs$ |
```

You are now in the course repository's root directory. All lab instructions will assume you are working from this location unless stated otherwise.

Step 7: Create and Activate dbt Virtual Environment

```
python3 -m venv dbt-venv
source dbt-venv/bin/activate
pip install "dbt-postgres==1.11.2"
ss -ltnp | grep 5432 || sudo service postgresql start
```



```
kofi@18ntiikxil02: ~/IT4065C-  ×  +  ▾
kofi@18ntiikxil02:~/IT4065C-Labs$ python3 -m venv dbt-venv
kofi@18ntiikxil02:~/IT4065C-Labs$ source dbt-venv/bin/activate
(dbt-venv) kofi@18ntiikxil02:~/IT4065C-Labs$ pip install dbt-postgres
Collecting dbt-postgres
  Downloading dbt_postgres-1.10.0-py3-none-any.whl (35 kB)
Collecting dbt-adapters<2.0,>=1.19.0
  Downloading dbt_adapters-1.22.4-py3-none-any.whl (172 kB)
    172.5/172.5 KB 1.9 MB/s eta 0:00:00
Collecting dbt-core>=1.8.0rc1
  Downloading dbt_core-1.11.2-py3-none-any.whl (1.0 MB)
    1.0/1.0 MB 4.4 MB/s eta 0:00:00
Collecting dbt-common<2.0,>=1.0.4
  Downloading dbt_common-1.37.2-py3-none-any.whl (87 kB)
    87.7/87.7 KB 5.4 MB/s eta 0:00:00
Collecting agate<2.0,>=1.0
  Downloading agate-1.14.0-py3-none-any.whl (95 kB)
    96.0/96.0 KB 4.9 MB/s eta 0:00:00
```

1. Verify installation:

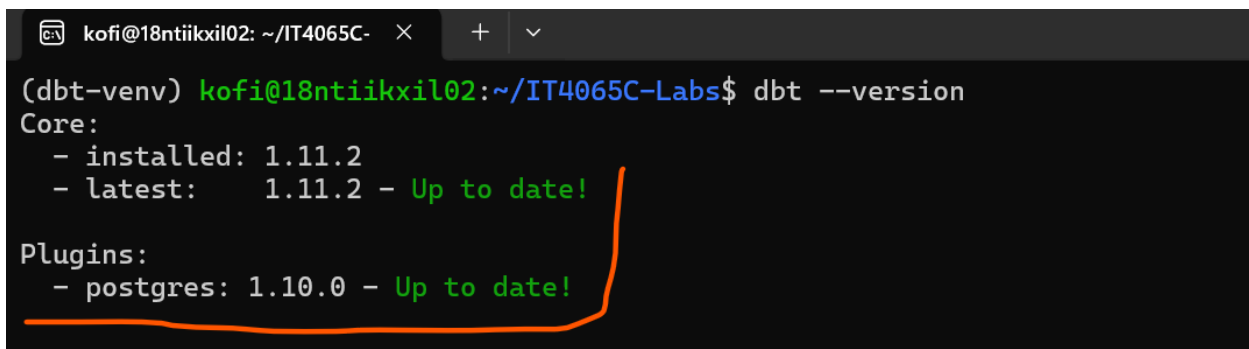
Run:

```
dbt --version
```

You must see the **Postgres adapter** listed.

Any time you work on dbt labs, you **must activate the virtual environment first**:

```
source ~/IT4065C-Labs/dbt-venv/bin/activate
```

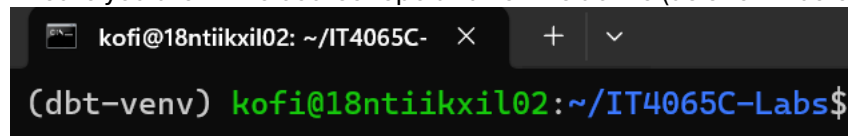


```
(dbt-venv) kofi@18ntiikxil02:~/IT4065C-Labs$ dbt --version
Core:
- installed: 1.11.2
- latest:    1.11.2 - Up to date!

Plugins:
- postgres: 1.10.0 - Up to date!
```

Step 8: Configure dbt profiles.yml (Project Profile)

1. Ensure you are in the course repo and venv is active (as shown below)



```
kofi@18ntiikxil02: ~/IT4065C-  ×  +  ▾
(dbt-venv) kofi@18ntiikxil02:~/IT4065C-Labs$
```

2. Navigate to the dbt project folder

```
cd dbt/it4065c_platform
```

```
kofi@18ntiikxil02: ~/IT4065C- x + v  
(dbt-venv) kofi@18ntiikxil02:~/IT4065C-Labs$ cd dbt/it4065c_platform  
(dbt-venv) kofi@18ntiikxil02:~/IT4065C-Labs/dbt/it4065c_platform$
```

3. Edit the project profile
 nano profiles.yml
4. Set the profile to this exact configuration (copy and paste)

```
it4065c_platform:  
  target: dev  
  outputs:  
    dev:  
      type: postgres  
      user: postgres  
      password: "Pa$$w0rd123!"  
      host: localhost  
      port: 5432  
      dbname: it4065c  
      schema: student_kofi  
      threads: 2  
      sslmode: prefer
```

Please
keep
as is

Note: Save changes and exit:

- **CTRL + O** → **Enter**
- **CTRL + X**

```
kofi@18ntiikxil02: ~/IT4065C- x + v - □ x  
GNU nano 6.2 profiles.yml *  
it4065c_platform:  
  target: dev  
  outputs:  
    dev:  
      type: postgres  
      user: postgres  
      password: "Pa$$w0rd123!"  
      host: localhost  
      port: 5432  
      dbname: it4065c  
      schema: student_kofi  
      threads: 2  
      sslmode: prefer  
^G Help ^O Write Out ^W Where Is ^K Cut  
^X Exit ^R Read File ^\ Replace ^U Paste
```

Note: This password was set earlier during database setup to allow dbt and psql to authenticate over localhost. Use it exactly as shown.

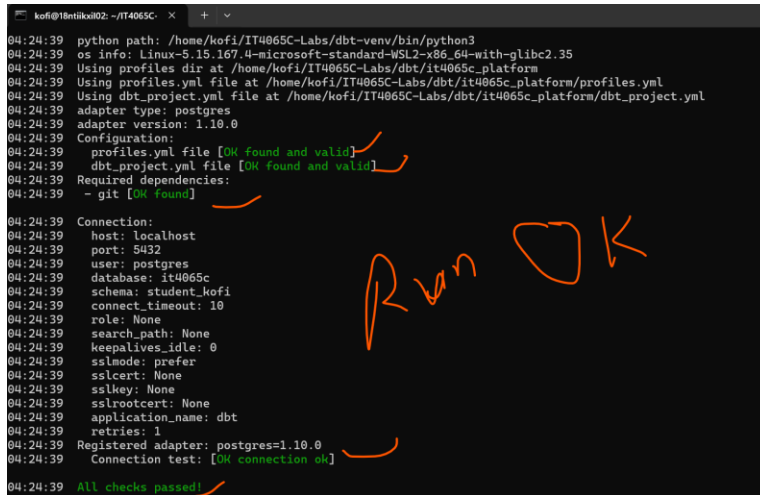
Step 9: dbt Debug

1. Run:
dbt debug

Success Criteria

You must see:

- Profile loaded successfully
- Connection test: OK



```
04:24:39 python path: /home/kofi/IT4065C-Labs/dbt-venv/bin/python3
04:24:39 os info: Linux-5.15.167.4-microsoft-standard-WSL2-x86_64-with-glibc2.35
04:24:39 Using profiles dir at /home/kofi/IT4065C-Labs/dbt/it4065c_platform
04:24:39 Using profiles.yml file at /home/kofi/IT4065C-Labs/dbt/it4065c_platform/profiles.yml
04:24:39 Using dbt_project.yml file at /home/kofi/IT4065C-Labs/dbt/it4065c_platform/dbt_project.yml
04:24:39 adapter type: postgres
04:24:39 adapter version: 1.10.0
04:24:39 Configuration:
04:24:39   profiles.yml file [OK found and valid]
04:24:39   dbt_project.yml file [OK found and valid]
04:24:39 Required dependencies:
04:24:39   - git [OK found]
04:24:39 Connection:
04:24:39   host: localhost
04:24:39   port: 5432
04:24:39   user: postgres
04:24:39   database: it4065c
04:24:39   schema: student_kofi
04:24:39   connect_timeout: 10
04:24:39   role: None
04:24:39   search_path: None
04:24:39   keepalives_idle: 0
04:24:39   sslmode: prefer
04:24:39   sslcert: None
04:24:39   sslkey: None
04:24:39   sslrootcert: None
04:24:39   application_name: dbt
04:24:39   retries: 1
04:24:39 Registered adapter: postgres=1.10.0
04:24:39 Connection test: [OK connection ok]
04:24:39 All checks passed!
```

If this fails, consult docs/troubleshooting.md.

📸 **Screenshot 2:** Take a screenshot showing successful dbt debug.

Step 10: Basic SQL Validation

1. Open PostgreSQL using the postgres user:

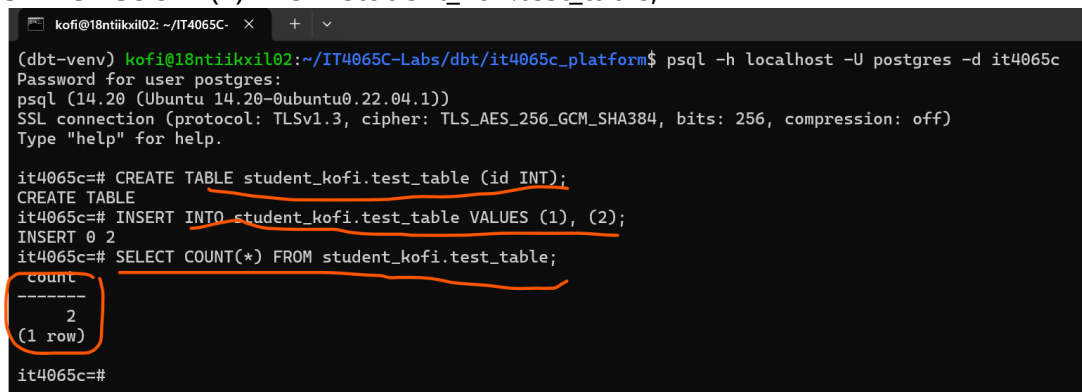
Run:

```
psql -h localhost -U postgres -d it4065c
```

Use password: Pa\$\$w0rd123!

2. Run:

```
CREATE TABLE student_kofi.test_table (id INT);
INSERT INTO student_kofi.test_table VALUES (1), (2);
SELECT COUNT(*) FROM student_kofi.test_table;
```



```
(dbt-venv) kofi@18ntiikx102:~/IT4065C-Labs/dbt/it4065c_platform$ psql -h localhost -U postgres -d it4065c
Password for user postgres:
psql (14.20 (Ubuntu 14.20-0ubuntu0.22.04.1))
SSL connection (protocol: TLSv1.3, cipher: TLS_AES_256_GCM_SHA384, bits: 256, compression: off)
Type "help" for help.

it4065c=# CREATE TABLE student_kofi.test_table (id INT);
CREATE TABLE
it4065c=# INSERT INTO student_kofi.test_table VALUES (1), (2);
INSERT 0 2
it4065c=# SELECT COUNT(*) FROM student_kofi.test_table;
 count
-----
      2
(1 row)

it4065c=#
```

📸 **Screenshot 3:** Take a screenshot showing the SELECT COUNT(*) result.

1. Exit PostgreSQL:
`\q`
-

Step 11: Lab 1 Deliverables

Submit **one MS Word or PDF file** on Canvas containing:

1. Screenshot 1: Repository structure (docs, labs, dbt)
2. Screenshot 2: Successful dbt debug output
3. Screenshot 3: SELECT COUNT(*) query output
4. Short reflection (3–5 sentences):

What schema am I responsible for, and why does schema isolation matter in governed data systems?

If you exit or reboot your VM, resume by:

1. Opening a terminal
`cd ~/IT4065C-Labs`
`source dbt-venv/bin/activate`